

# Tianhao(Tony) Ye

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## EDUCATION

### Columbia University

Master of Science in Operations Research, GPA: 3.60

New York, NY

Aug. 2025 – Dec. 2026

- **Coursework:** Financial Engineering, Machine Learning, Quantitative Risk Management, Data-Driven Methods in Finance

### Boston University

Bachelor of Arts in Economics and Mathematics, **Cum Laude**

Boston, MA

May. 2025

## SKILLS

**Technical Skills:** Python(pandas, NumPy, scikit-learn, PyTorch), Excel, R, Tableau, MySQL, Bloomberg, Linux

**Professional Skills:** Factor Research, VaR/Expected Shortfall, Machine Learning, Deep Learning, Time Series Modeling, Stress Testing, Scenario Analysis, Optimization, Volatility Analysis, Sensitivity Analysis, Barra Risk Model, Out-of-Sample Validation

## WORK EXPERIENCE

### Columbia University, IEOR Department

Teaching Assistant, IEOR 4525: Machine Learning for Financial Engineering & Operations Research

New York, NY

Sep. 2026 – Dec. 2026

- Develop Python Jupyter notebooks and lead office hours covering linear models, decision trees, and deep neural networks.

### Mingshi Investment Management

Quantitative Research Intern

Shanghai, China

May. 2026 –Aug. 2026

- Engineered alpha factors across intraday market microstructure, financial statements, and analyst forecasts covering A-share universe; after industry and MSCI Barra CNE6 risk factor neutralization, identified 10+ factors with low correlation to the existing factor library and annual cross-sectional Rank IC  $\geq 3.5\%$  against forward VWAP returns.
- Backtested factors across multiple holding periods via Rank IC, IC stability, quantile long-short portfolios, and turnover-adjusted sharpe ratio with realistic round-trip costs; top factors reached a theoretical long-short Sharpe ratio of 2.5+.
- Designed an event-time IC framework for fundamental features and operator transformations, separating announcement impulse from post-event drift and correcting systematic one-day alignment error via filing-timestamp forensics.
- Developed and benchmarked PIT, embargoed walk-forward logistic regression, XGBoost, LightGBM, and XGBoost-logistic stacking models using 70+ signals interacted with lagged macro variables; the best achieved 7.5% OOS Barra-neutral Rank IC (+60% vs. the strongest standalone factor) and 1.5 IR versus CSI 1000 after costs and trading constraints.

### Shanghai River Fund Management Co., Ltd.

Credit Risk Intern

Shanghai, China

Dec. 2024 – Feb. 2025

- Conducted timely credit analysis for internal rating assignments on Diversified Industries counterparties, evaluating financial condition and creditworthiness, improving rating turnaround efficiency by 25% for ongoing risk monitoring.
- Prepared credit review reports for Risk Team approval, including financial spreading and detailed analysis of clients' 10-K/10-Q filings, helping inform internal capital requirements and improving credit review consistency by 18%.
- Performed credit underwriting on borrowers through industry research, review of transaction memos and credit agreements, and Excel-based EBITDA bridge and capital structure analysis to assess repayment capacity and default risk.
- Executed stress testing and scenario analysis across multiple portfolios by applying price volatility and interest rate shocks, estimating position-level P&L under hypothetical and historical scenarios to identify downside exposures.

### Newbridge Securities Corporation

Market Risk Intern

New York, NY

May. 2024 – Jul. 2024

- Produced daily risk reports for the Fixed Income desk and developed a Python preprocessing pipeline for yield-curve and duration analysis, reducing report production time by 30% and improving timeliness of risk delivery.
- Evaluated market risk model outputs using time-series validation, VaR backtesting, and P&L attribution analysis; redesigned the control workflow in VBA to improve consistency and reduce manual intervention.
- Conducted stress testing and scenario analysis across multiple portfolios by applying interest rate and price volatility shocks, translating hypothetical and historical market moves into position-level P&L to identify concentrated risk exposures.
- Managed a customized risk data mart covering PnL, NAV, and position data, and automated multi-source ingestion, cleansing, and uploads through Python and MySQL, reducing manual data pulls by 20% and improving refresh timeliness.

## PROJECTS EXPERIENCE

### Machine Learning Project – Loan Default Prediction

Risk Analyst & Team Lead

New York, NY

Jan. 2026 – Apr. 2026

- Built a Python/scikit-learn pipeline to predict charge-offs using 1.6M consumer loan records, screening 150 raw fields for availability at loan origination and combining missing-value imputation, feature scaling, and dimensionality reduction.
- Compared logistic regression, random forest, and k-nearest-neighbor classifiers through cross-validated grid search; selected logistic regression, achieving 0.69 AUROC on the most recent 10% holdout and identifying interest rate, term, FICO score, and debt-to-income ratio as key risk drivers.